An Evaluation of Technology to Mitigate Freon R-218, Carbon Disulfide, and Carbonyl Sulfide Contamination to Support Sabatier Integrated Operations

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The Four-Bed Carbon Dioxide Removal (FBCO2) hardware is used on the International Space Station (ISS) to remove metabolic carbon dioxide (CO_2) from the cabin. Currently, the CO_2 is vented overboard. However, the ISS Program will be integrating a Sabatier Carbon Dioxide Reduction system in ~2026. The Sabatier recovers oxygen from CO_2 to reduce resupply of water and oxygen from the ground. A Sabatier reactor, designed by Collins Aerospace, operated on ISS from 2010 to 2017 before the catalyst degraded due to a combination of contaminants including fluorine, chlorine, and sulfur. While the origin of these contaminants could not be conclusively determined, they are suspected of originating from contaminants on ISS. Freon R-218 has been measured at concentrations on ISS between 1500 and 4500 mg/m^3. Low levels of carbon disulfide (CS_2) and carbonyl sulfide (COS) have also been detected. While these concentrations do not pose a risk to crew health, CO_2 removal in FBCO2 is performed using materials that co-adsorb Freon R-218, COS, and CS_2. These contaminants have been measured at elevated concentrations in the CO_2 product from FBCO2. Without mitigation, these contaminants are expected to decompose in the Sabatier and ultimately foul the catalyst. To mitigate this risk with the new Sabatier hardware, an evaluation of eight technology approaches to protect the Sabatier reactor from R-218, COS, and CS_2 was conducted. An overview of technologies considered, criteria for comparison, and the results of a simplified trade study are presented. Additionally, follow-on work to finalize a technology decision is discussed

Acronyms and Nomenclature

AC	=	Activated Carbon
AC-TSAC	=	Air-Cooled Temperature Swing Adsorption Compressor
CO_2	=	carbon dioxide
Comp	=	compressor

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COS = carbonyl sulfide
CS_2 = carbon disulfide
D = diameter
$DMSO_2$ = dimethylsulfone
DMSD = dimethylsilanediol
FBCO2 = Four-Bed Carbon Dioxide Removal
HF = hydrogen fluoride
IL = ionic liquid
ISS = International Space Station
L = length
MOF = Metal-organic framework
QD = Quick Disconnect
PNNL = Pacific Northwest National Laboratory
R-218 = Freon Refrigerant-218, octafluoropropane
SA = Sabatier Assembly
TBD = to be determined
TRL = Technology Readiness Level
W = watts

I. Introduction

IN 2009, Hamilton Sundstrand (now Collins Aerospace) delivered the Sabatier Assembly (SA) to the International Space Station (ISS) [1]. The SA employed a Sabatier reactor to recovery oxygen (in the form of water) from metabolic carbon dioxide (CO_2). After activation in 2010, the SA operated intermittently until October 2017 when operation was discontinued due to performance degradation. The unit was returned to the ground for Test, Teardown, and Evaluation and analysis confirmed contaminants had reduced catalyst activity by 70%. The primary contaminants found in the catalyst were sulfur, silicon, fluorine, and chlorine. The analysis determined that sulfur and silicon were the most detrimental poisons with less, though measurable, impact from chlorine and fluorine. Sulfur and silicon were attributed to dimethylsulfone ($DMSO_2$) and volatile methyl siloxanes from upstream sources, respectively. Fluorine was attributed to a degraded Teflon seal downstream of the reactor and a source of chlorine could not be identified [2].

Based on the findings and recommendations from the SA analyses, the ISS Program initiated activities to design a SA 2.0 as part of an exploration-class Environmental Control and Life Support System demonstration on ISS. Specific upgrades of the SA 2.0, relative to the original SA, included an updated design that would allow for maintenance closer to the component level rather than the orbital replacement unit level, and efforts to improve reliability by making the system more tolerant to contamination [3]. In particular, the SA 2.0 design included a 'getter' sorbent bed using HSS11, an activated carbon with a Hamilton Sundstrand-developed proprietary treatment for targeted performance, following performance testing with four molecules to mimic anticipated contaminants. These included $DMSO_2$ to introduce sulfur, trimethylsilanol and decamethylcyclopentasiloxane to introduce silanol/siloxanes, and dichloromethane to introduce chlorine. No fluorine compound was added based on the assumption that chlorine and fluorine would behave similarly in the catalyst and could both be represented by the chlorine contaminant. The results of testing informed the getter design combined with contaminant load assumptions based on historical testing. Freon Refrigerant-218 (R-218, octafluoropropane) was considered as part of the design load at 175 mg/m^3. The resulting getter bed was baselined for SA 2.0 and design efforts proceeded [4].

A second component of the exploration-class ECLSS system is the Four Bed CO_2 (FBCO2) removal system which will be integrated upstream of the SA 2.0. On-orbit sampling of the CO_2 product from FBCO2 demonstrated four new contaminants of potential concern for the Sabatier catalyst: Refrigerant-218 (R-218), isobutane, carbon disulfide (CS_2), and carbonyl sulfide (COS) [5]. These new contaminants are introduced due to their adsorption on zeolite 13X, the primary CO_2 removal sorbent in FBCO2. Ground testing of a FBCO2 engineering unit confirmed that these contaminants were concentrated in the product CO_2 that would be delivered to SA 2.0 [5] and posed three potential issues. First, while the getter was designed to adsorb sulfur from $DMSO_2$, neither CS_2 nor COS were tested with the HSS11 sorbent. Second, HSS11 was not tested for R-218 adsorption. The assumed load the getter would experience was 175 mg/m^3. However, on-orbit sampling showed levels as high as 4500 mg/m^3. Additionally, due to the high operating temperature of the Sabatier reactor, there was concern that R-218 could break down and produce hydrogen fluoride (HF) which would be expected to cause severe corrosion downstream of the reactor. Third, isobutane was not tested in the getter and no analysis had been done to predict its effect on the system.

Due to the apparent risk to long-duration SA 2.0 operation caused by the newly identified contaminants, two efforts were initiated. The first effort sought to evaluate the effects of R-218, COS, and CS_2 on Collins Aerospace Sabatier catalyst. The results of that activity are reported elsewhere [6]. The second effort, reported here, sought to identify solutions that would remove or eliminate the contaminants before they reached the Sabatier reactor.

II. Mitigation Solution Analysis and Design Constraints

All potential solutions were constrained by the SA 2.0 design envelope and feasible integration points available in March 2024. Figure 1 provides a simplified overview of the integration considerations. Note that, while shown

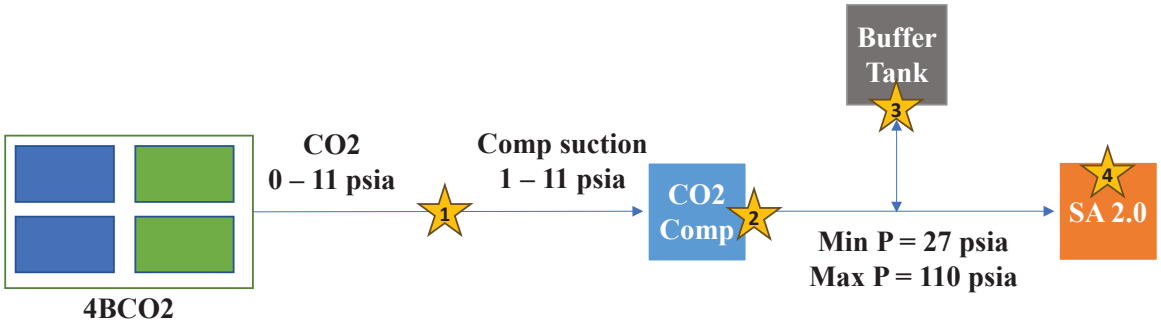


Figure 1. Integration options for mitigation technologies. *Stars indicate locations where mitigation solutions could feasibly be implemented based on hardware design. Getter and flow controller are both located internal to SA 2.0.*

separately in the figure, the CO_2 compressor (Comp) is part of the SA 2.0 envelope. The FBCO2 provides CO_2 at outlet pressures between 0 and 11 psia (0-75.8 kPa) depending on the timing of the half-cycle. The SA 2.0 CO_2 compressor provides suction at 1-11 psia (7-75.8 kPa). A quick disconnect (QD) located between 4BCO2 and the compressor (at star #1) could conceivably allow for integration of a mitigation technology between the two. Similarly, downstream of the outlet to the CO_2 compressor is another QD (star #2) where a mitigation technology coud be incorporated. The compressor can provide up to 110 psia (limited by the maximum design pressure of the buffer tank) to the buffer tank and the rest of the SA 2.0 hardware. The buffer tank has a series of QDs at its inlet (star #3). Notably, the CO_2 flow controller upstream of the Sabatier reactor has a minimum pressure requirement of 27 psia and the Sabatier operates at a nominal pressure of ~12.5 psia. This provides the bounding pressures for integration at either of the two locations. Finally, the getter is located just upstream of the Sabatier reactor (but downstream of the flow controller). Volume constraints will not permit additional hardware to be added within the design, but a direct replacement of the getter would be conceivable within SA 2.0 (star #4) if an alternative option could be identified to meet the new requirements in addition to those of the existing getter [4].

An initial evaluation of the getter redesign (for implementation at star #4) was conducted using the same models from the original design but with added contaminants isobutane, CS_2, and COS, and increased R-218 concentration [7]. The analysis sought to estimate the time until breakthrough of each contaminant. Table 1 shows the results of the analysis. The adsorption potential denotes the potential energy of a molecule as it approaches a surface for adsorption. Higher values indicate that adsorption is less energetically favorable while lower values indicate adsorption is more energetically favorable. The estimated life of the bed is based on the updated load concentrations and an assumption of a 2 inch (5.08 cm) diameter x 6 inch (15.24 cm) long bed as described previously [4].

Table 1. Analysis of getter redesign for SA 2.0 to accommodate updated contaminant loads.

Contaminant	Poison	Load (mg/m3)	Ads. Potential (K-mol/mL)	Capacity (mg/g)	Estimated Life (hrs)	Notes
$DMSO_2$	S	1.10	22.2	9.3	399	
DMSD	Si	0.08	22.4	6.3	3801	
Isobutane	C	5.82	80.9	-	0	Ads. on AC unlikely
CS_2	S	0.63	32.6	0.3	9.9	Weakly adsorbed
COS	S	0.05	111.9	-	0	Ads. on AC unlikely
R-218	F	1008	10.8	168.9	12.8	Will displace weak adsorbers over time

As shown, isobutane and COS are unlikely to adsorb on activated carbon (AC) as evidenced by high adsorption potential. Isobutane, as a hydrocarbon, could conceivably break down in the Sabatier reactor, and produce carbon dioxide and carbon monoxide. However, given the high concentration of hydrogen, would ultimately react to form additional methane. Previous testing with other hydrocarbons showed no carbon formation in a Sabatier reactor with identical catalyst [8]. The inability of the getter to adsorb COS and only weakly adsorb CS_2 suggests that a different getter material would be required or a second method of mitigating the sulfur contributors will be needed. The getter was designed for $DMSO_2$ and dimethylsilanediol (DMSD) and the estimated life is consistent with anticipated performance. DMSD was selected over the previous analysis based on samples acquired from ISS. DMSD was observed in samples while trimethylsilanol and decamethylcyclopentasiloxane were not. Notably, if the getter was used to capture CS_2 and R-218, the bed would have to be replaced every 9.9 (based on CS_2) to 12.8 hrs (based on R-218). This replacement schedule would be prohibitive for any mission and a solution within SA 2.0 (star #4) was omitted from further consideration.

Based on the other three potential integration locations (stars #1, 2, and 3 in Figure 1), three general approaches were considered: sorbent-based mitigation, cryocondensation, and incineration. While one technology solution each was identified for the cryocondensation and incineration approaches, a total of seven technologies in six architecture solution combinations were identified for sorbent-based mitigation. Table 2 provides a list and description of the architecture options. 'Consumable' beds refer to sorbent beds that must be replaced after reaching adsorption capacity. 'Regenerable' beds refer to sorbent beds that can regenerated and used cyclically under controlled conditions.

III. Solution Criteria and Weights

While multiple potential technology solutions were identified for R-218 and sulfide mitigation, the ultimate goal was implementation into a flight system in tandem with SA 2.0 flight and activation on ISS (targeted for April 2026). This required criteria by which to compare technologies for suitability. The ISS program-defined criteria included:

- R-218 Mitigation: the solution should effectively remove or eliminate R-218 from the CO_2 feed stream.
- COS/CS_2 Mitigation: the solution should effectively remove or eliminate COS and CS_2 from the CO_2 feed stream.
- Integration Complexity: the solution should minimize complexity of integration with FBCO2 and SA 2.0
- Power: the solution should minimize required power

Table 2. Potential architecture solutions for R-218 and sulfide mitigation.

Architecture #	Technology Description
1	Regenerable Norit Activated Carbon (AC) Bed, with and without separate consumable sulfide sorbent bed
2	Regenerable Metal Organic Framework (MOF) Beds for R-218 removal, with consumable sulfide sorbent bed
3	Consumable AC bed for R-218 removal, with sulfide sorbent bed
4	Regenerable AC with Ionic Liquid Beds for R-218 removal, with consumable sulfide sorbent bed
5	Membrane-Supported Ionic Liquid, engineered to preferentially adsorb R-218, combined with consumable sulfide sorbent bed
6	Air-Cooled Temperature Swing Adsorption Compressor (AC-TSAC), Existing technology resized and operationally modified for R-218 and sulfide removal
7	Crycooler Condensation, Existing technology resized and operationally modified for R-218 and sulfide removal
8	Plasma Incinerator, Existing technology repurposed for R-218 mitigation, combined with consumable acid gas/sulfide sorbent bed

- Volume: the solution should minimize required volume
- Technology Readiness Level (TRL): the solution should be of sufficiently high TRL to have flight hardware within 2 years.

After establishing the criteria, four cases were considered. Case 1 sought to identify the 'best technical solution' to mitigation R-218, COS, and CS_2. In this case, the solution was not constrained by integration complexity, power, or volume. This allowed an evaluation of the technology based purely on effectiveness of mitigating the contaminants. Case 2 sought to identify the solution that met all the criteria described above. Case 3 sought to identify a solution that addressed only R-218 and Case 4 sought to identify a solution that addressed only COS and CS_2. Cases 3 and 4 were included in the event it was determined that one or more of the contaminants either did not affect the Sabatier catalyst or did not degrade the catalyst such that the 3 year life requirement could not be met [6]. Under those circumstances, there was potential that the program would accept the risk of additional contamination with no mitigation.

Based on these case definitions, weights of 0 to 5 were given to each criterion based on relative importance from the program perspective. A score of zero indicated the criteria had no bearing on the decision for the corresponding case. A score of 5 indicated that the criteria was of highest importance when selected a technology for the corresponding criteria. Weights between 0 and 5 were assigned by project managers based on relative importance to criteria scored a 5 and relative to other criteria. Table 3 shows the final breakdown of criteria and corresponding weights for each case. Note for power that technologies requiring less than 250W would be weighted '1.' However, for technologies requiring more that 250W of power, they were weighted at '3.' In all cases, contaminant mitigation and TRL level were weighted highest. Volume was determined to be the next most important for flight cases, with integration complexity and power having the lowest weight.

Table 3. Selection criteria and weights by Case.

Criteria	Case 1 Weight	Case 2 Weight	Case 3 Weight	Case 4 Weight
R218 mitigation	5	5	5	0
COS/CS2 mitigation	5	5	0	5
Integration Complexity	0	3		
Power	0	1 (<250W*), 3 (>250W)		
Volume	0	4		
TRL Level	5	5		

*250W is the power required to operate the compressor from the original Sabatier.

IV. Architecture and Technology Descriptions

Nine unique technologies, described below, were considered for the architectures outlined in Table 2.

A. Regenerable AC R-218 Beds

Architecture #1 assumed dual regenerable AC beds potentially combined with a consumable sulfide bed, as shown in Figure 2. A single AC bed would operate at any given time while the other would be temperature and/or pressure regenerated. Norit RB3 was recommended by Norit for use in this application. The material is commercially available as a 3mm diameter x 6 mm length low dust pellet. Based on its chemistry, it is also possible that Norit RB3 could capture COS and CS_2, but no data is yet available to confirm. Should testing confirm capture of COS and CS_2, no separate sulfide bed would be required in the architecture. Integration could be completed either upstream (star #1) or downstream of the CO_2 compressor (star #2). The final location would depend on the performance of the AC at the available pressures.

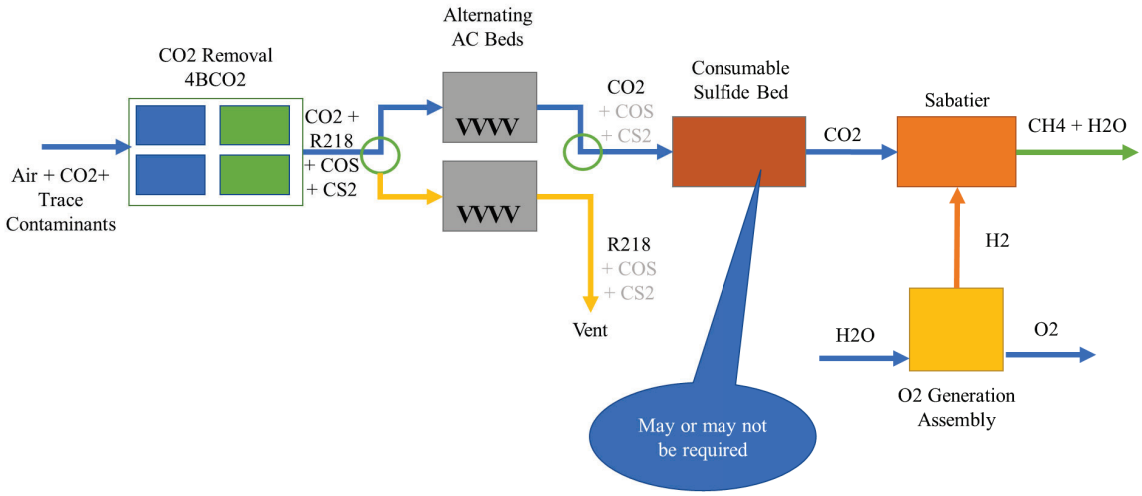


Figure 2. Regenerable AC R-218 beds with possible consumable sulfide bed. *Molecule names in grey indicate they may or may not be present depending on the performance of the AC.*

B. Regenerable R-218-Adsorbing Metal Organic Framework (MOF)

Architecture #2 assumed dual thermal/vacuum regenerable MOF beds combined with a consumable sulfide bed, as shown in Figure 3. Like the Regenerable AC R-218 bed architecture, a single MOF bed would operate at any given

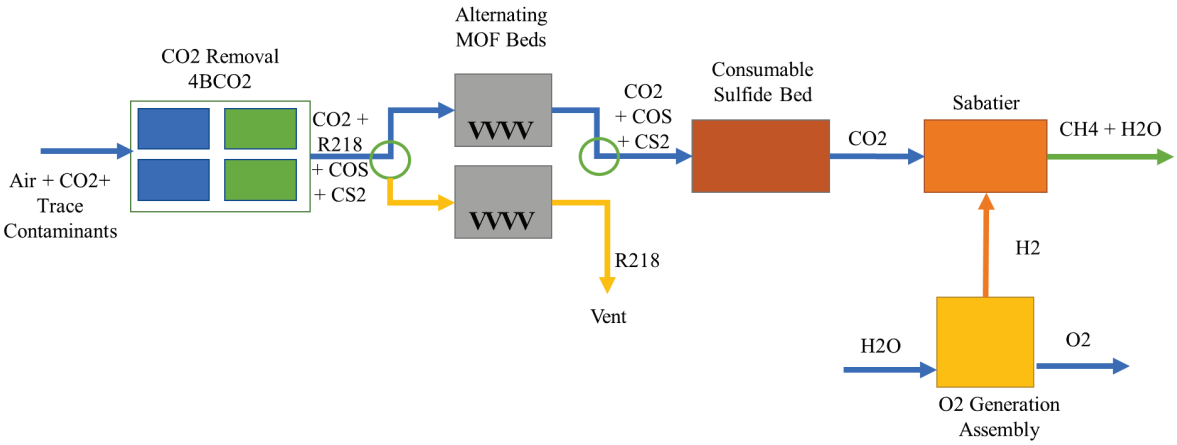


Figure 3. Regenerable MOF sorbents and consumable sulfide bed.

time while the other would be regenerated. A team at Pacific Northwest National Laboratory (PNNL) have developed a number of engineered MOFs with high "fluorophilic" activity, meaning that they preferentially adsorb fluorinated molecules [9, 10, 11, 12, 13, 14, 15, 16, 17]. Additionally, the PNNL group has demonstrated both pelletization and sieving into required grain sizes for various applications. While none of the available MOFs are specific for R-218, the team postulated that manipulating the density of the fluorophilic sites in the porous structure would result in 'superfluorophilicity' to improve R-218 selectivity over CO_2 and water. Additionally, the maximum production capacity of MOFs at the PNNL laboratory is 1-2 grams. Testing of the capability and subsequent implementation into a flight system would require 1 kg or more, necessitating a secondary manufacturer after proof of performance at a lower scale. Integration could be completed either upstream (star #1) or downstream of the CO_2 compressor (star #2). The final location would depend on the performance of the AC at the available pressures.

C. Consumable AC R-218 Beds

Consumable AC beds for R-218 capture were included in Architecture #3. The beds assumed the same catalyst used in the getter of SA 2.0 [4], but with modified geometry to allow for less frequent replacement. Calculations of the consumable bed were performed with an assumption of R-218 capacity of 100mg R-218 per gram of activated carbon and a load rate of 1260 mg/h (at R-218 ISS worst case concentration of 4500 mg/m^3) [7]. Table 4 provides the volume of AC required for 7 and 30 day replacements and the bed dimensions assuming consistent length (L)/diameter (D) ratio of 3:1.

Table 4. Consumable AC Bed Sizing for R-218.

Replacement Frequency	Volume AC Required	Bed Dimensions (L/D = 3)
7 days	3.85 L	4.6 in D, 13.9 in L (11.8cm D, 35.3cm L)
30 days	16.50 L	7.5 in D, 22.6 in L (19cm D, 57.4cm L)

D. Consumable Sulfide Beds

Architectures containing Regenerable MOF beds, Consumable AC beds, Regenerable AC with IL, Membrane-supported IL, Plasma Incineration, and potentially Regenerable AC for R-218 all require a consumable sulfide bed for implementation. The Norit RB3 AC is one potential option for sulfide capture and may also be regenerable. However, there are a number of other commercially available sorbents specifically designed to adsorb sulfides. Because of their broad use industrially, consumable sulfide beds were assumed to be low risk for technology implementation and no further analysis or research was conducted to choose a specific solution for this trade effort.

E. AC-TSAC

The AC-TSAC was originally designed to be an alternative to the mechanical CO_2 compressor in a Sabatier system. AC-TSAC has the benefit of having no moving parts, relying purely on thermal swing to regenerate sorbent beds and to pressurize product CO_2 for the Sabatier [18, 19]. Architecture #6 assumed a modified AC-TSAC design, notionally shown in Figure 4, in which zeolites 4A and 5A would be used in lieu of the original design plan of zeolite 13X. Zeolites function through size exclusion into their pores. Zeolites 4A, 5A, and 13X have pore sizes of 4 Angstroms (A), 5A, and 10A, respectively. The kinetic diameters of CO_2 and R-218 are 3.2A and 6.3A, respectively. Accordingly, CO_2 can fit into the 4A and 5A pores, but R-218 would be too large. Therefore, CO_2 would be expected to be captured by AC-TSAC, allowing R-218 to be discarded. The CO_2 would then be thermally regenerated from the bed and fed to SA 2.0. AC-TSAC architecture is shown in Figure 5. This approach has the added benefit of eliminating a piece of hardware from SA 2.0 by replacing the mechanical CO_2 compressor and CO_2 storage

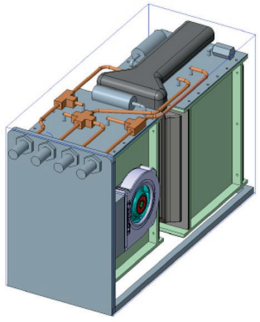


Figure 4. Notional AC-TSAC bed replacing mechanical CO_2 compressor volume.

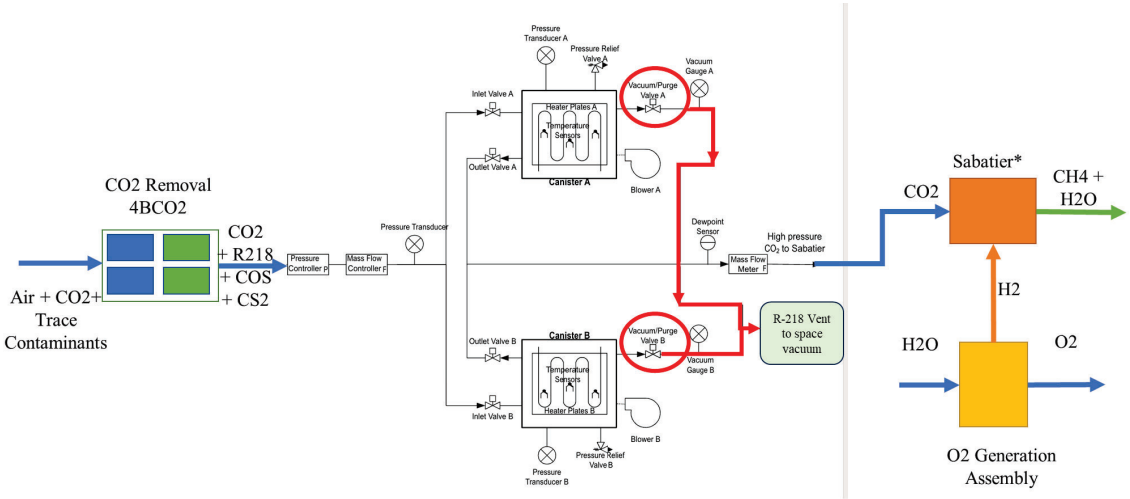


Figure 5. AC-TSAC replacing CO_2 compressor.

tank. However, zeolites 4A and 5A are likely to adsorb COS and CS_2, which would result in these contaminants feeding the Sabatier with CO_2. If testing shows that the sulfur degrades the Sabatier catalyst, an alternative or additional sorbent will be needed.

F. Cryocooler condensation

CO_2 Deposition is a technology developed as an alternative to FBCO2 to remove CO_2 from cabin air [20]. The technology uses cryocoolers with a finhead cold tip to cryogenically cool and deposit CO_2 in a semi-batch process. This effectively separates the CO_2 from the bulk air flow on one head, while the other warms and vaporizes the captured CO_2 product. By alternating cooling and warming, continuous removal can be achieved. Architecture #7 assumed the use of a modified cryocooling approach. As shown in Figure 6, the condensation temperature of CO_2 is 194K. The condensation temperatures of CS_2, COS, and R-218 are 161K, 134K, and 90K, respectively. By maintaining the cryocooler temperature between 161K and 194K, the approach can separate pure CO_2 while allowing the contaminants to remain in the gas phase to be vented or combined with fresh feed for further refinement. At an operating temperature of 185K, the vapor pressures of the CS_2, COS, and R-218 are well above their partial pressures, indicating they would remain in the vapor phase throughout the process. However, operation of the finhead at this temperature, would only condense ~57% of the CO_2 in a single batch. Repeated batch processes with fresh feed would allow for continued CO_2 separation until either the fin head reaches maximum capacity, or the partial pressure of R-218 (limiting contaminant) reaches its vapor pressure. Cryocooler condensation architecture is shown in Figure 7.

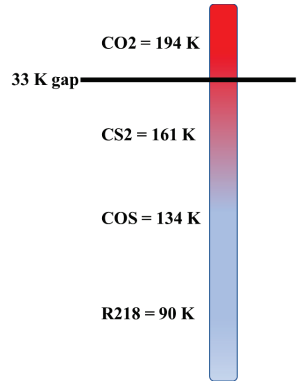


Figure 6. Condensation temperatures of CO_2 and contaminants.

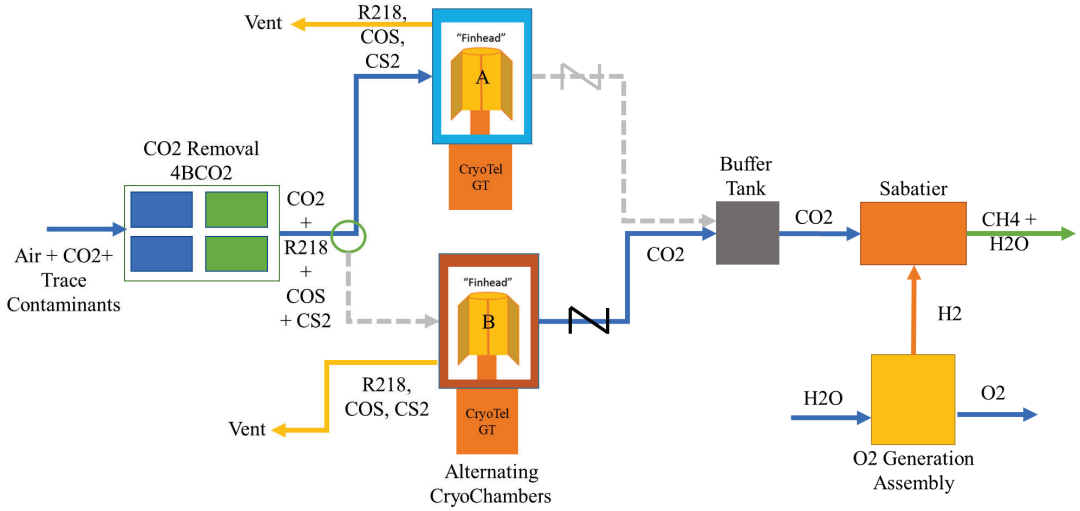


Figure 7. Cryocooler condensation.

G. AC with Supported Ionic Liquid

Previous work by NASA with the University of Nevada, Las Vegas sought to develop a sorbent to remove DMSD for the water purification assembly [21]. The same strategy was used to explore the feasibility of a designer ionic liquid (IL) supported on an activated carbon for R-218 removal in Architecture #4. Ten relevant studies were reviewed [22, 23, 24, 25, 26, 27, 28, 29, 30, 31] looking at ILs and Deep Eutectic Solvent adsorption of refrigerants and fluorocarbons. Typical Henry constants and liquid solubility data for R-218 in various solvents was identified and solubility was consistently on the order of $x_{R\text{-}218} = 0.025$ (mole fraction). One study measured solubility data of three fluorinated gases, including R-218 in four combinations of IL cations and anions [24]. One of the more promising results was $[C_8H_4F_{13}C_1Im][NTf_2]$ reporting that the IL could adsorb 1.115% of its weight in R-218. This equates to ~11.5 mg R-218/g of IL when exposed to a pure stream of R-218. Adsorption from a mixed stream would likely

experience reduced efficiency due to mass transport limitations and competitive adsorption with other contaminants and humidity. Compared to solid sorbents, which have an order of magnitude higher capacity, these results suggest very large systems would be required to implement the technology solution.

H. Membrane Supported Liquid Membrane

Architecture #5 was similar to Regenerable AC with IL but using a membrane-supported IL. The initial thought was that if a more electronegative molecule, such as R-218, could transport through membrane pores more readily than the carrier gas (i.e., CO_2), then the membrane approach might be feasible. Hirayama *et. al* [32] evaluated the permeation properties of 19 gases through silicone rubber membrane, a PEO* membrane, and a PF/PEO† membrane. The reported permeability coefficients of CO_2 at 25 °C were 1600, 200, and 66 Barrer, respectively. As a comparison, the reported permeability coefficients of R-218 at 25 °C were 108, 0.158, and 0.101 Barrer, respectively. As can be seen, the significantly lower permeability coefficients of the R-218 relative to CO_2 indicate that the CO_2 will be much more readily transported across the membranes than R-218.

I. Incineration

For terrestrial use, R-218 is primarily recommended for recycling [33]. Disposal is conducted through a combination of high temperature incineration, followed by discharge in a counterflow scrubber for the acid gases produced during incineration. NASA has developed a plasma-based technology for managing volatile organic compounds off gassing from thermal waste processors as part of the "Trash-to-Gas" project [34]. The hardware uses spark plugs inserted radially along the length of the reaction chamber to induce a plasma. Initial studies with Freon-113 showed that plasma incineration was possible and rapid with the hardware. Architecture #8 assumed repurposing the plasma generator, shown in Figure 8, as a plasma incinerator for R-218. Due to the known acid gas (hydrofluoric acid (HF)) decomposition products of R-218, the architecture assumed a downstream acid gas sorbent bed as shown in Figure 9. Notably the intentional production of HF introduces challenges for materials of construction that will not be susceptible to corrosion and the added challenge of containment for crew safety.

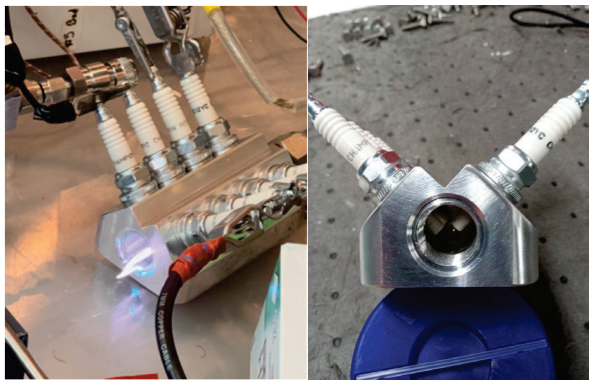


Figure 8. Plasma incinerator with (left) and without (right) active flow and plasma.

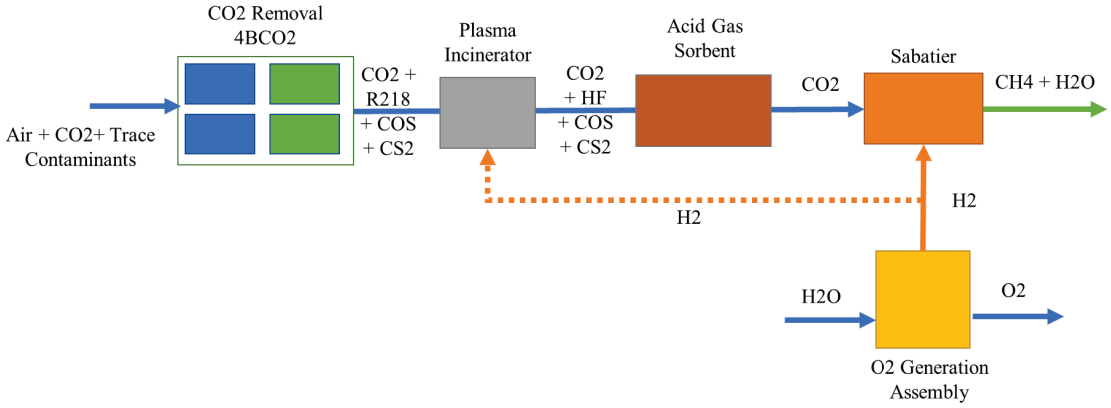


Figure 9. Plasma Incineration and Acid Gas Capture.

* PEO in this case was a co-polymer of methoxy-terminated poly(ethylene glycol) methacrylate (MEMA) and poly(ethylene glycol) dimethacrylate (EDMA)

† PF/PEO here refers to a membrane blend of poly (ethylene oxide) and polyvinylidene fluoride.

V. Trade Overview

Each architecture was evaluated per the criteria and weights described above. Critical assessments of each option were made which resulted in three early eliminations. Specific information for each criteria was collected for the remaining five options.

A. Criteria Assessments and Technical Challenges

1. *Early Eliminations*
    a. *Consumable AC R-218 Bed + Sulfide Sorbent Bed*
    Analysis showed that a consumable AC bed, as sized for SA 2.0 would saturate in ~ 6 hours. Even sized larger, a consumable bed would require excessive consumables over a 3 year mission. The architecture was deemed non-feasible and eliminated from further consideration.
    b. *Consumable or Regenerable Bed with AC-Supported Ionic Liquid*
    Analysis of the AC-supported ionic liquid showed that the estimated adsorption capacity for R-218 would be an order of magnitude lower than simple AC. Additionally, a designer IL would require significant development and scaling of material synthesis to be feasible. The architecture was deemed non-feasible and eliminated from further consideration.
    c. *Membrane Supported Ionic Liquid*
    Analysis of membrane-supported IL showed a fundamental issue with the approach. Larger, more electronegative molecules such as R-218 would not preferentially pass through a membrane versus smaller carrier gases (e.g. CO_2). The architecture was deemed non-feasible and eliminated from further consideration.

2. *Plasma Incineration + Consumable Acid Gas Bed*
    Table 5 shows the results of the criteria assessment for the incineration approach.

Table 5. Criteria assessment for Plasma Incinerator.

Option	R218 Mitigation	COS/CS2 Mitigation	Integration Complexity	Power	Volume	TRL
Incineration + Consumable Acid Gas Bed	Expected, but needs to be demonstrated and optimized.	Yes, with acid gas bed	Integrated at star #1. No changes to either sub-system.	80-100W	Single locker	2

Technical challenges identified for this architecture included testing, analysis, and development required before implementation. First, there is a concern the plasma could convert CO_2 to oxygen and carbon monoxide. Due to the high energy release of oxygen mixing with hydrogen in the Sabatier reactor, there may safety concerns. Testing would be needed to determine if R-218 can be incinerated without producing oxygen and carbon monoxide (or recombining before feed to the Sabatier). Another possibility is that oxygen and carbon monoxide are produced, but then react and produce other products that may prove easier to manage than HF. Analysis would be needed to determine whether SA 2.0 materials are compatible with acid gases, and if not, redesign would be required. Development would also be needed with this approach. First, development and/or process optimization would be needed to achieve 100% R-218 conversion in a single pass flow through. Second, careful design, and potentially hardware development would be needed to ensure safe operation in the presence of oxygen, hydrogen, and acid gases. The incinerator is at a TRL 2. Acid gas beds are simple sorbent beds, widely used, and can be considered TRL 9.

3. Regenerable MOF Beds + Consumable Sulfide Sorbent Bed

Table 6 shows the results of the criteria assessment for the regenerable MOF beds approach.

Table 6. Criteria assessment for Regenerable MOF Beds.

Option	R218 Mitigation	COS/CS2 Mitigation	Integration Complexity	Power	Volume	TRL
Regenerable MOF Beds + Consumable Sulfide Sorbent Bed	Expected, but needs to be engineered, demonstrated, and optimized.	Yes.	Option 1: Integrated at star #1. No changes to either sub-system. Option 2: Integrated at star #2 to achieve higher partial pressure for adsorption.	?	?	2

The key challenge with the MOF beds would be the development of the MOF itself. Development would require multiple formulations and iterations, competitive adsorption testing, pelletization, and pellet testing before bed design could even begin. Notably the MOF is at a TRL 2. However, swing bed design is well understood and widely used in ISS life support systems. This means that the hardware design is high TRL.

4. AC-TSAC

Table 7 shows the results of the criteria assessment for the AC-TSAC approach.

Table 7. Criteria assessment for AC-TSAC.

Option	R218 Mitigation	COS/CS2 Mitigation	Integration Complexity	Power	Volume	TRL
AC-TSAC	Yes	TBD KSC testing, 4A and 5A expected to capture both COS and CS2	Eliminates mechanical CO2 compressor and tanks from SA 2.0. Added complexity because requires Collins design change. Reduced complexity because replaces moving parts and potential sources of contamination.	300W Ave, 800W Peak	CO2 Compressor + Avionics	3-4

The current AC-TSAC has never been operated to maintain delivery pressure at the 27 psia required by the Sabatier flow controller. Design modifications would be required to achieve the pressure and testing would be required to confirm the operational capability. There is a risk that COS and/or CS_2 will co-adsorb with CO_2 and be delivered to the Sabatier. This may necessitate an alternative material be developed or a downstream high-pressure sulfide bed be included in the architecture. The TRL is estimated at between 3 and 4. However, if new sorbent is needed, the TRL will drop accordingly.

5. Cryocooler Condensation

Table 8 shows the results of the criteria assessment for the cryocondensation approach.

Table 8. Criteria assessment for Cryocooler condensation.

Option	R218 Mitigation	COS/CS2 Mitigation	Integration Complexity	Power	Volume	TRL
Cryocooler Condensation	Yes	Yes	Eliminates SwRI compressor but requires one tank for CO2 buffer. Can provide up to 45 psia CO2.	200W, 180W to MTL	CO2 Compressor + Avionics	3-4

Cryocoolers are relatively simple hardware that have been used in other applications on ISS. While this architecture has the fewest unknowns of any of the technologies, there are several challenges that would need to be addressed before implementation. First, testing would be required to confirm that CS_2 is at a sufficiently low concentration to prevent liquefaction at the target temperature. Second, testing would be required to confirm thermal control within the 33K band between CO_2 and CS_2 condensation temperatures. The cryocoolers and cooling fins would replace the

mechanical CO_2 compressor in the architecture, but the storage tank would need to be either modified to allow flow-through to the Sabatier, or an alternative tank would be required to allow for flow buffering. This is because the cryocoolers cannot control the flow rate of CO_2 during the vaporization process. Ideally, a cryocondensation system would match the FBCO2 cycle time (60 minutes of CO_2 with 80 minute total cycle). Demonstration and possible optimization would be required to ensure high concentrations of CO_2. Finally, the system would need to be demonstrated at sub-ambient pressures since all testing to-date has been at slightly positive pressure. This is not expected to be a concern due to operation of these cryocoolers at pressures as low as 8mbar in other applications.

6. Regenerable AC R-218 Bed + Consumable Sulfide Sorbent Bed

Table 9 shows the results of the criteria assessment for the regenerable AC bed approach. Norit RB3 promises to provide a solution for capturing R-218 and possibly COS and CS_2. Challenges with this approach include demonstrating the capability of the material to capture the contaminants and regenerate in a reasonably sized and infrequently cycled bed. There are three integration location options for the architecture.

For integration at star #1, testing would be required to confirm there is sufficient suction from the mechanical CO_2 compressor to pull gas through the bed. Integration at either star #2 or star #3 would require near-term design modifications to SA 2.0 to implement.

Table 9. Criteria assessment for Regenerable AC R-218 Bed.

Option	R218 Mitigation	COS/CS2 Mitigation	Integration Complexity	Power	Volume	TRL
Regenerable AC Beds + Consumable Sulfide Sorbent Bed	Yes	Yes	Option 1 = At star #1, lower pressure gives one design Option 2 = At star #2, higher pressure gives smaller design (near term design change for SA 2.0 to accommodate) Option 3 = At star #3, with separate inlet/outlet from tank.	<200W Peak	For sulfide adsorption values (based on amines in literature) – 200-300 mL	3-4

B. Trade Results

Criteria weights for four cases were evaluated. In the end, all cases yielded the same ranking of architectures, despite differences in weighting criteria. The architectures, rankings, and justifications for Cases 1 (Best Technical Solution) and 2 (Full Removal of R-218, COS, and CS_2) are shown in Table 10. The same information for Cases 3 (R-218 Removal Only) and 4 (COS and CS_2 Removal Only) are shown in Table 11.

Despite these rankings, near term activities are needed to make a final decision on a mitigation technology:

- Rank 1 Recommendation: Regenerable AC Bed Architectures
  - Analysis of NORIT RB3 is necessary to demonstrate the capacity, selectivity, and regenerability of the material. This data is needed before the Regenerable AC for R-218 architecture can be confirmed.
- Rank 2 Recommendations: Cryocooler Condensation Architectures
  - Three activities are needed before Cryocooler condensation can be selected. First, testing is needed to evaluate the total CO_2 deposited during a 60 minute half cycle (to coincide with FBCO2 operation). Second, testing is needed to determine if the finhead temperature can be controlled within the targeted range to selectively deposit CO_2. Third, testing is needed to confirm that at the target temperatures, CS_2 is not liquified on the finhead.
- Rank 3 Recommendation: AC-TSAC Architectures
  - Testing is needed to demonstrate sorbent selectivity for CO_2 versus the contaminants. Testing of additional sorbents is warranted if 4A and 5A adsorb significant quantities of COS and/or CS_2.

No near-term recommendations are made for Regenerable MOF beds or Plasma Incineration. These approaches may be appropriate for longer-term consideration. However, given the timeline specified during this effort, these are unlikely to produce near-term solutions and should be prioritized lower for additional work.

Table 10. Case 1 and 2 Rankings and Justification.

Architecture #: Description	Case 1 Ranking Best Technical	Case 2 Ranking Full Removal R218, COS, CS2
1: Regen AC Beds + Non-regen S Beds	1 – assumes NORIT RB3 can remove R-218, COS, and CS2, assumes lower power than cryocondensation because adsorption cycle is long, and infrequently regenerated	1 – relatively small volume hit, but simplifies integration if at star #1. assumes NORIT can remove R-218, COS, and CS2, assumes lower power than cryocondensation because adsorption cycle is long, and infrequently regenerated
7: Cryocooler Condensation	2 – least unknowns, but similar capability to rank 1, higher Power	2 - least unknowns, but similar capability to rank 1, lower power than AC-TSAC with similar sizing and integration
6: AC-TSAC	3 – assumes sorbent does not adsorb R218 or sulfur compounds	3 – assumes sorbent does not adsorb R218 or sulfur compounds, higher power than cryocoolers with similar sizing and integration
2: MOF Regen Beds + S beds	4 - high risk on TRL, starting from scratch in MOF development	4 - high risk on TRL, starting from scratch in MOF development
8: Incineration + acid gas sorbent beds	5 – low TRL, but demonstrated breakdown of freons, still need to deal with acid gases (sorbent beds), and produces HF which must be captured.	5 – low TRL, but demonstrated breakdown of freons, still need to deal with acid gases (sorbent beds) and produces HF which must be captured.

Table 11. Case 3 and 4 Rankings and Justification.

Architecture #: Description	Case 3 Ranking Only R218 Removal	Architecture #: Description	Case 4 Ranking Only COS & CS2 Removal
1: Regen AC Beds	1 – relatively small volume hit, but simplifies integration if at star #1, assumes NORIT RB3 can remove R-218, assumes lower power than cryocondensation because adsorption cycle is long, and infrequently regenerated	1: Non-regenerable S beds	1 – Simple replaceable sulfur bed
7: Cryocooler Condensation	2 – least unknowns, but similar capability to rank 1, lower power than AC-TSAC with similar sizing and integration	7: Cryocooler Condensation	2 - least unknowns, but similar capability to 1, higher Power
6: AC-TSAC	3 – assumes sorbent does not adsorb R218, higher power than cryocoolers with similar sizing and integration	6: AC-TSAC	3 – assumes sorbent does not adsorb sulfur compounds
2: MOF Regen Beds	4 - high risk on TRL, starting from scratch in MOF development	2: Non-regenerable S beds	N/A - Same solution as above when MOF beds are removed from architecture.
8: Incineration + acid gas sorbent beds	5 – low TRL, but demonstrated breakdown of freons, still need to deal with acid gases (sorbent beds), and produces HF which must be captured.	8: Incineration + acid gas sorbent beds	5 – low TRL, but demonstrated breakdown of freons, still need to deal with acid gases (sorbent beds)

VI. Conclusions and Forward Work

Eight architectures were evaluated to mitigate R-218, COS, and CS_2 contamination in the CO_2 feed to SA 2.0. Three were eliminated before applying trade criteria and weights due to a general lack of feasibility with the approaches. The remaining five were ranked based on four Cases. The results were identical in each case, though for

slightly different reasons in some instances. Recommendations were made for near-term testing to address known challenges with each architecture and help the program make a final decision

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